

Spatial Analysis of Colorado Zillow Housing Listings

Mapping, regression, nearest-neighbor analysis, clustering, and interactive Leaflet visualization

123

cleaned listings

44

cities represented

82

ZIP codes

Project Objective

Use spatial methods to understand where listings are located and how prices vary.

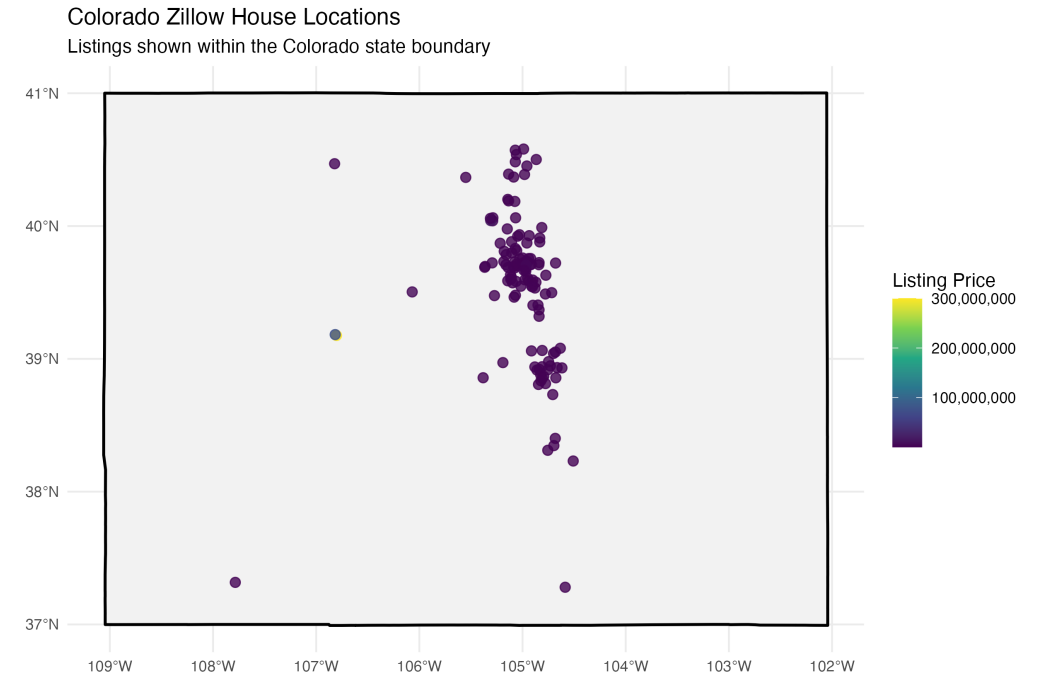
Treat Zillow listings as spatial points using latitude and longitude.

Map listings within Colorado state and county polygon boundaries.

Use regression to evaluate how price relates to housing characteristics and location.

Use nearest-neighbor and cluster analysis to assess whether listings are random or geographically concentrated.

Create an interactive Leaflet map for exploring individual listings.



Dataset and Preparation

The data were cleaned before mapping, modeling, and spatial analysis.

Metric	Value
Number of listings	123
Number of cities	44
Number of ZIP codes	82
Minimum price	\$49,999
Median price	\$635,000
Mean price	\$3,972,322
Maximum price	\$300,000,000
Median area square feet	2,540
Median price per square foot	\$254

Duplicate listing IDs were removed.

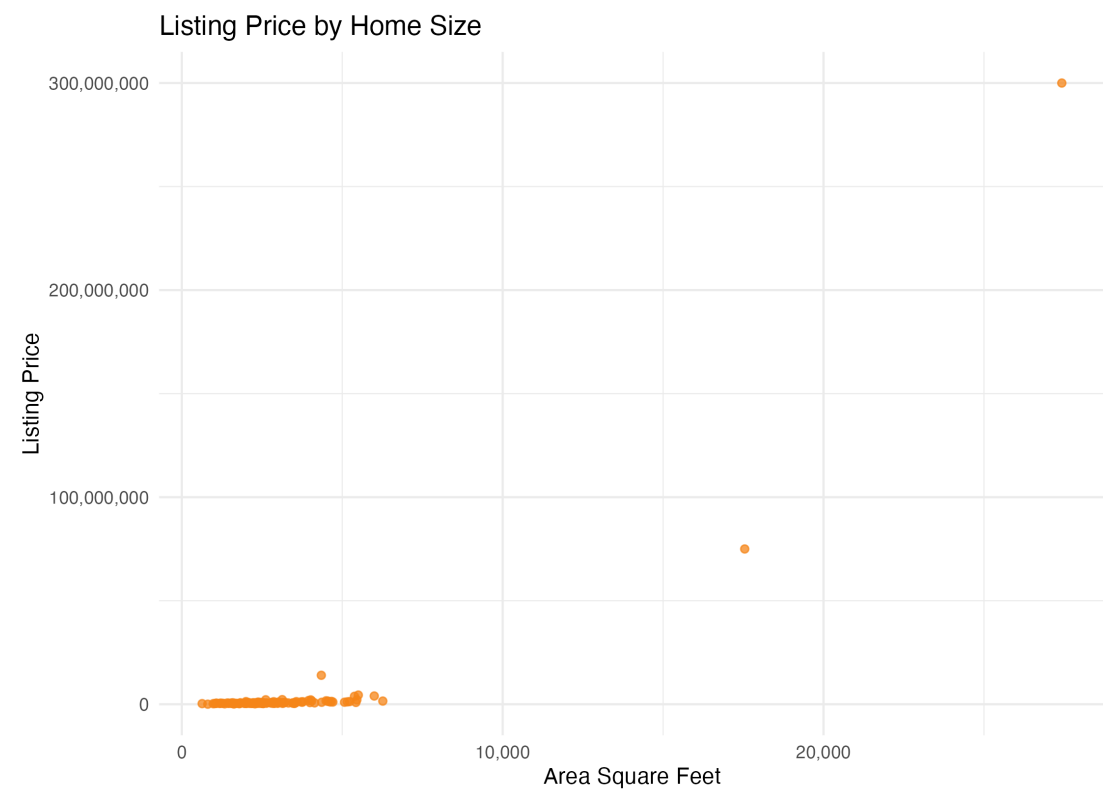
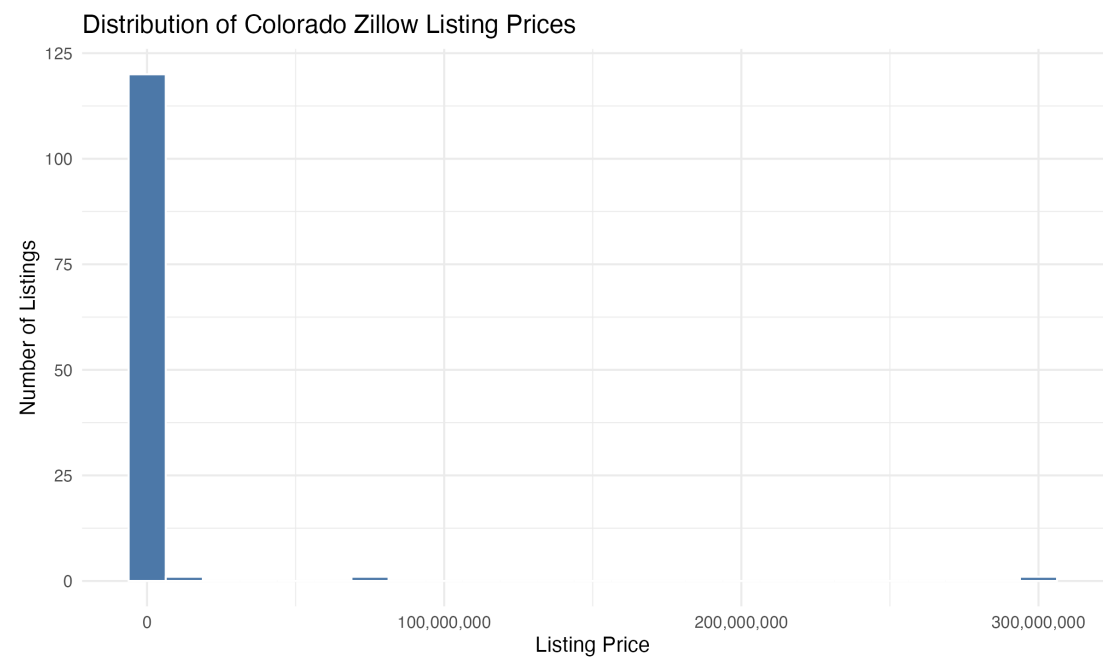
Records with missing price, coordinates, beds, baths, or square footage were excluded.

Price per square foot and log price were created for analysis.

Coordinates were checked to keep listings inside a reasonable Colorado range.

Exploratory Summary: Price Is Highly Skewed

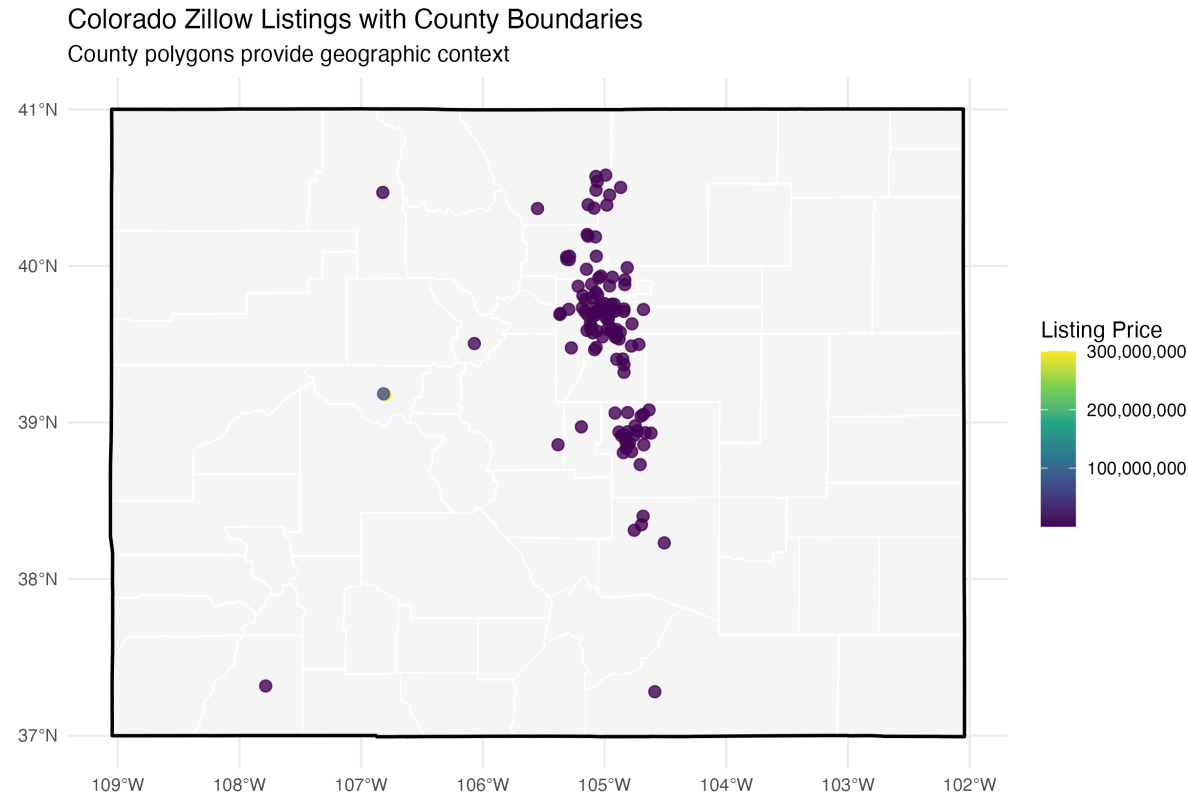
A small number of luxury listings strongly affects the mean.



Most listings fall in moderate price ranges, but a few very high-value homes create a long right tail. This is why log listing price was used in the regression model.

Listings Are Concentrated Along the Front Range

The map shows a strong geographic concentration rather than an even statewide pattern.



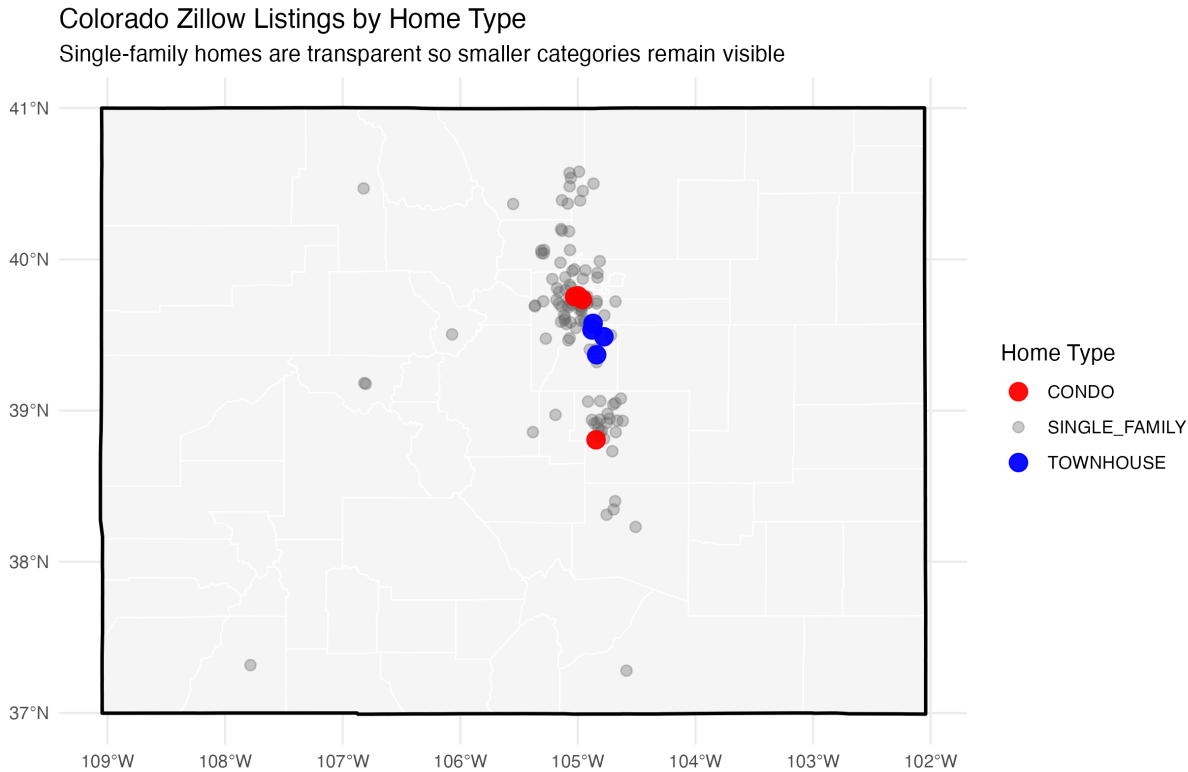
Listings are concentrated near Denver, Colorado Springs, Boulder, and surrounding communities.

County polygons provide geographic context for the listing points.

The spatial pattern suggests housing data should be interpreted geographically, not only numerically.

Home Type Pattern

Single-family homes dominate the sample, so smaller groups should be interpreted cautiously.



Home Type	Listings	Median Price	Median Area
Single-family	113	\$650,000	2,583
Condo	5	\$375,000	1,201
Townhouse	5	\$585,000	1,780

The transparent single-family layer keeps condos and townhouses visible.

Small condo and townhouse counts mean their summary statistics are less stable.

Home type was included as a predictor in the regression model.

Regression Analysis

Log listing price was modeled using property characteristics and location.

Variable	Estimate	p-value
Beds	-0.020	0.6751
Baths	0.111	0.0432
Area Sq Ft	0.000195	<0.001
Single Family	0.442	0.0486
Townhouse	0.435	0.1498
Days on Zillow	-0.00477	0.0001
Latitude	0.163	0.0236
Longitude	-0.437	<0.001

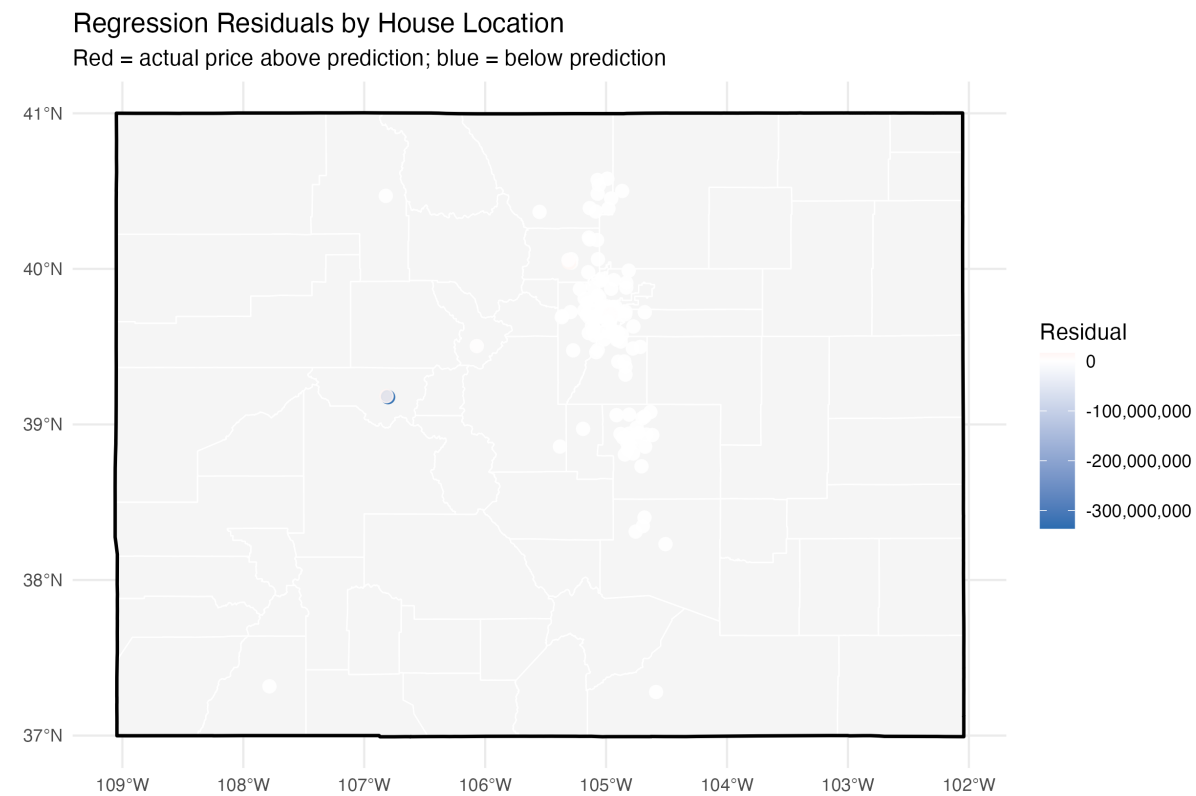
Square footage, bathrooms, days on Zillow, latitude, and longitude were meaningful predictors.

The location coefficients show that geography remained important even after adding property characteristics.

Using log price helped reduce the influence of extreme listing prices.

Regression Residuals by Location

Residual mapping shows where the model overpredicts or underpredicts price geographically.



Positive residuals mean actual price was higher than predicted.

Negative residuals mean actual price was lower than predicted.

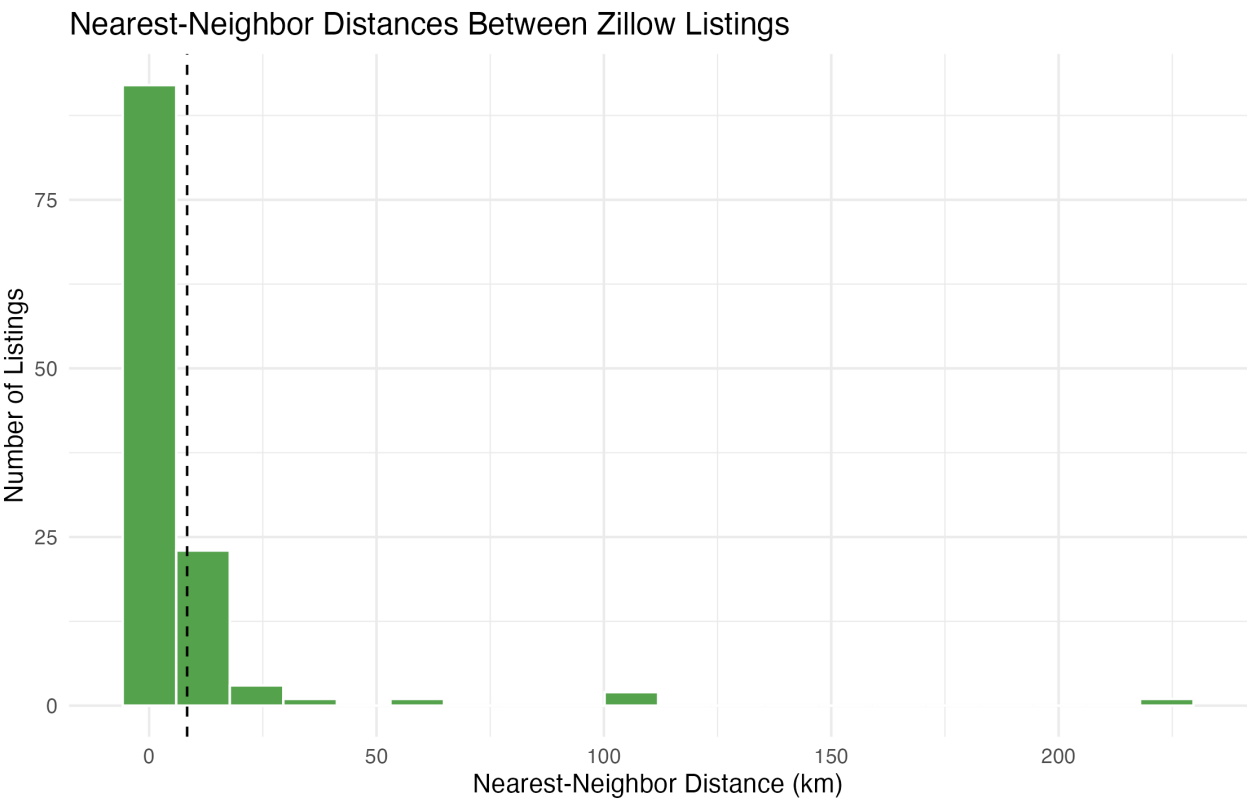
Large residuals may reflect factors not in the model, such as views, neighborhood quality, school zones, or luxury amenities.

Nearest-Neighbor Analysis

Listings are closer together than expected under complete spatial randomness.

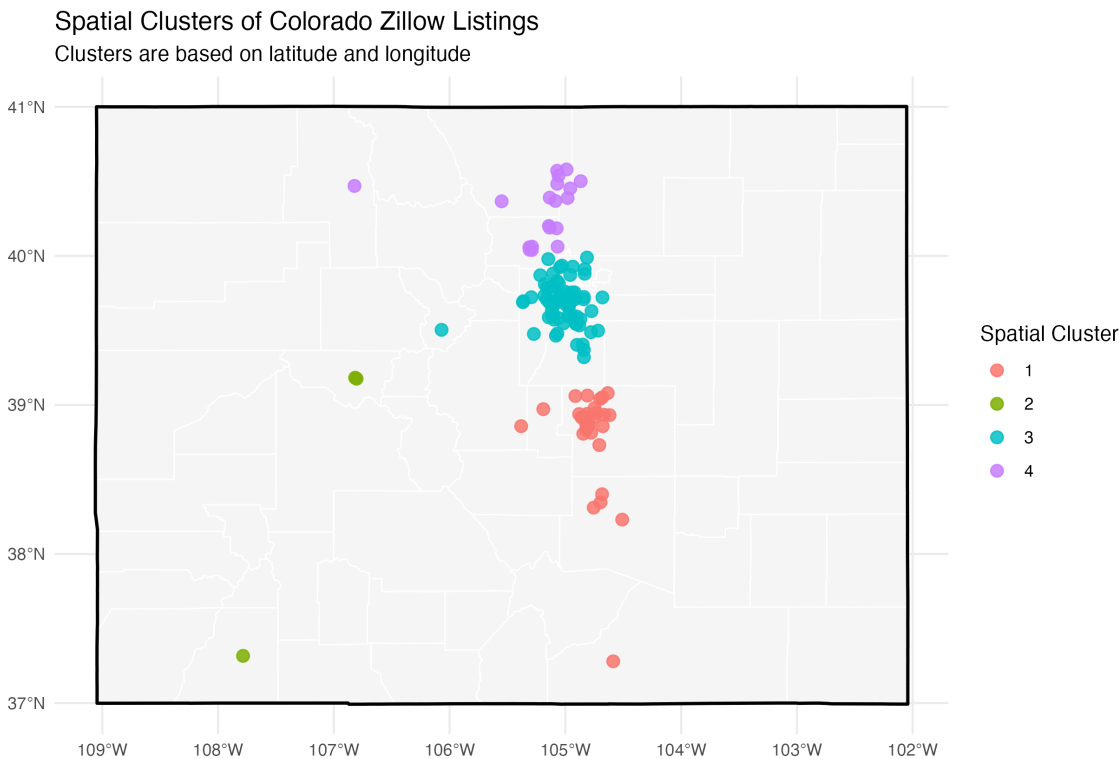
Metric	Value
Listings	123
Study area	81,832.085 sq km
Observed mean NN distance	8.356 km
Expected mean NN distance	12.897 km
Nearest-neighbor index	0.648
Interpretation	Clustered

Because the index is less than 1, the listings are spatially clustered rather than randomly distributed.



Spatial Cluster Analysis

Coordinate-based clusters summarize regional listing groups.



Cluster	Count	Median Price	Approx. Location
I	29	\$460,000	Southern CO / Colorado Springs
II	3	\$75,000,000	Mountain / luxury listings
III	72	\$662,500	Denver & central Front Range
IV	19	\$650,000	Northern Front Range / Boulder

The largest cluster is around Denver and the central Front Range.

The small luxury cluster has very high prices but only three listings.

Clusters help connect price patterns to regional geography.

Interactive Leaflet Map

The final HTML map allows viewers to inspect individual listings.

Shows Colorado boundary, county polygons, and Zillow listing points.

Popups include address, city, actual price, predicted price, residual, home type, beds, baths, area, and cluster.

Submitted separately as a zipped HTML map so it can be opened in a web browser.

Submitted file

**colorado_zillow_interactive_map_SU
BMIT.zip**

Unzip, then open
colorado_zillow_complete_interactive_map.html

Conclusion

Spatial analysis added insight that tables alone could not show.

0.648

nearest-neighbor index

72

largest cluster

\$635K

median listing price

\$3.97M

mean listing price

Colorado Zillow listings in this sample are clustered, especially along the Front Range.

Location remains important after accounting for home characteristics in regression.

Residual and cluster maps help identify geographic patterns that are not obvious from tables.

The project demonstrates how mapping, modeling, and interactive visualization can work together in spatial housing analysis.